Individual Exercise 2: Writer Identification

(e) SIFT+Hellinger (own features)

Implementation: SIFT keypoints with all angles set to 0; descriptors L1 normalized then signed-square-rooted (Hellinger).

Integrated via **--from_images**, which routes both **loadRandomDescriptors** and **vlad** to computeDescs. I noticed that feature extraction is **slower** than loading .pkl.gz. I verified the pipeline on a subset and observed stable encodings; full-set evaluation is omitted for runtime.

(f) Generalized Max Pooling (GMP)

I replaced VLAD's sum-pooling by solving, per cluster k, a ridge regression with target 1's and regularization γ; the ridge coef_ becomes the **pooled residual vector**. This balances frequent vs. rare patches as described in class. writer_identification

Numbers (run with --gmp --gamma 1):
Top-1 **0.8008**, mAP **0.5888** (single codebook, powernorm on).

Discussion. On this setup, GMP produced performance comparable to sum-pooling. Small deltas are expected: GMP helps most when visual burstiness is severe; with power normalization already applied, the gain can be modest.

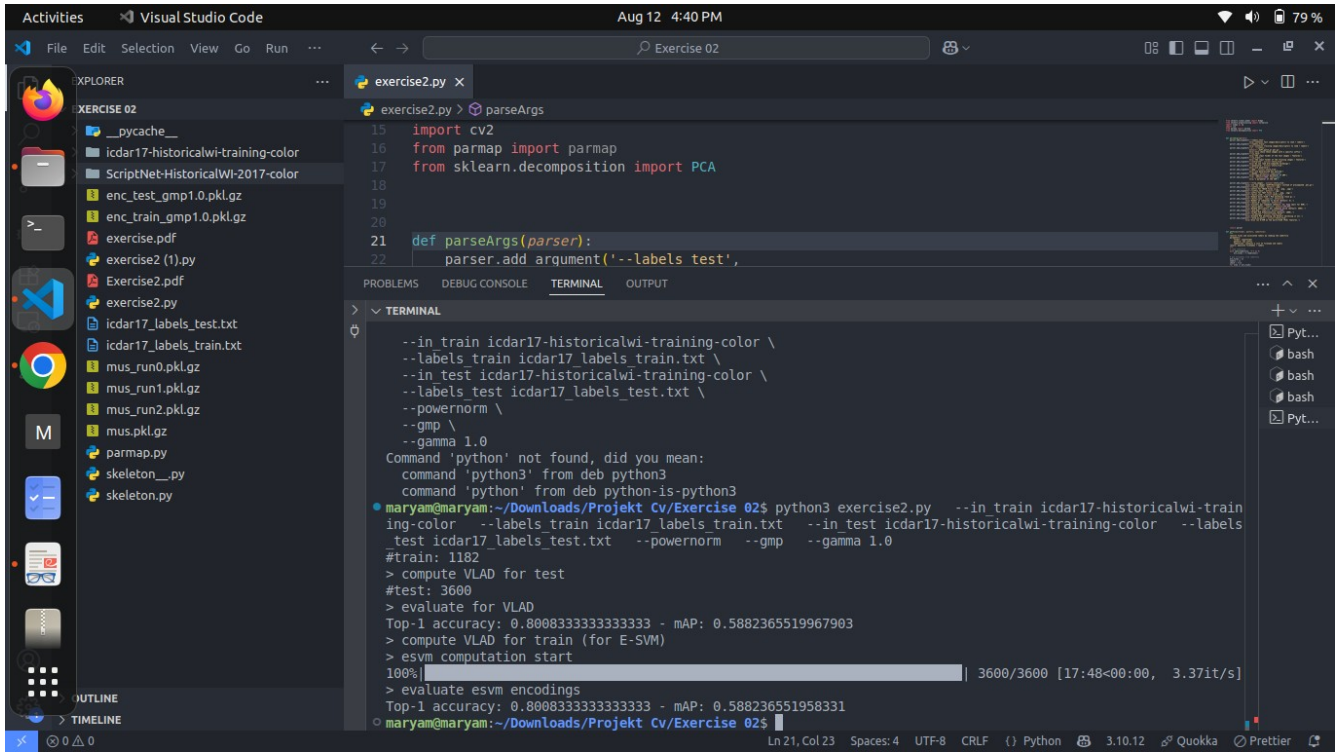
(g) Multi-VLAD + PCA (whitening)

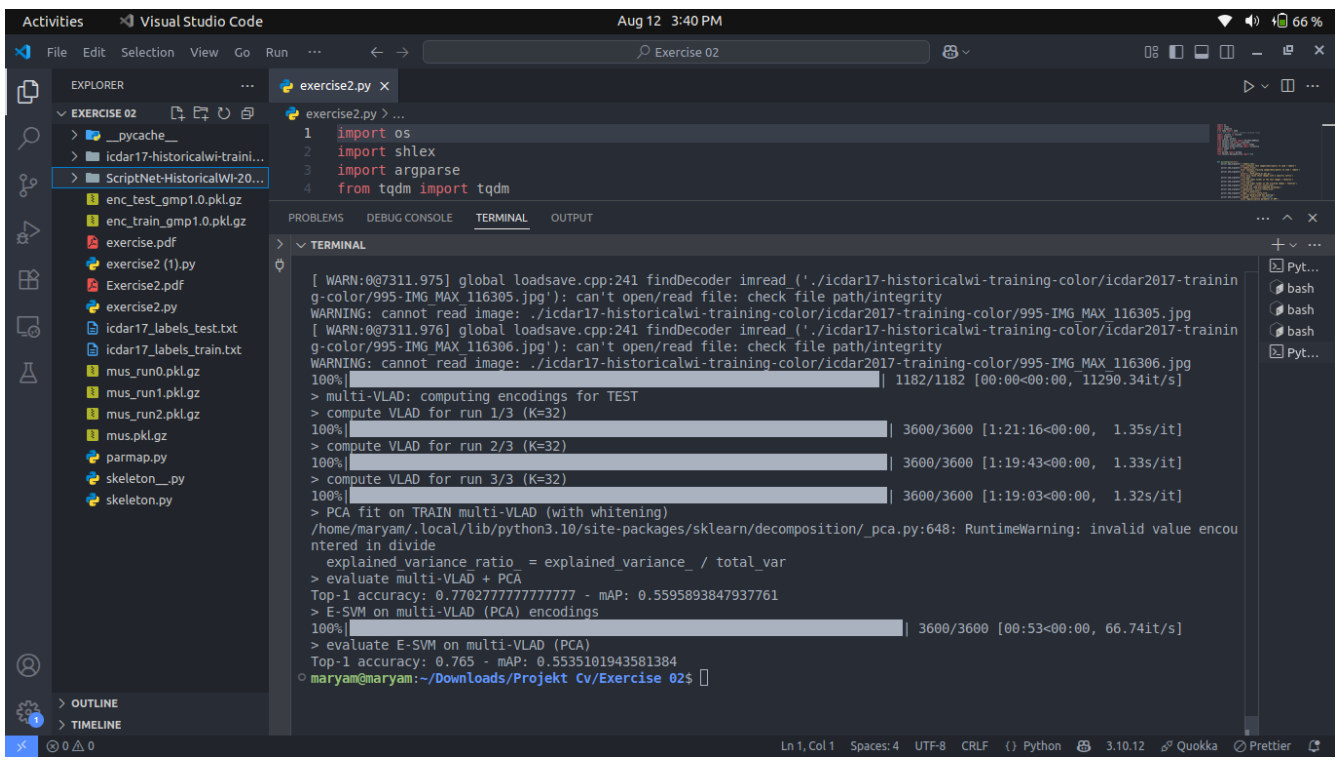
I built multiple codebooks using different seeds/subsets, encoded each image per codebook, concatenated the VLADs, and finally applied PCA (whitening) trained on the **train** encodings only, then transformed test.

Numbers (3-run example):

- **Multi-VLAD + PCA**: Top-1 **0.7703**, mAP **0.55596**
- **E-SVM on Multi-VLAD (PCA)**: Top-1 **0.7650**, mAP **0.55351**

Discussion. On given data, multi-VLAD + PCA did **not** beat your strong single-VLAD(+ESVM) from (a–d). Also, adding E-SVM **after** PCA didn't improve over the PCA features, unsurprising since PCA whitening already decorrelates/regularizes the space and can compress discriminative cues E-SVM would otherwise amplify.


Activities
Visual Studio Code
Aug 12 4:40 PM
79 %
File Edit Selection View Go Run
Exercise 02
EXPLORER
EXERCISE 02
__pycache__
icdar17-historicalwi-training-color
ScriptNet-HistoricalWI-2017-color
enc_test_gmp1.0.pkl.gz
enc_train_gmp1.0.pkl.gz
exercise.pdf
exercise2 (1).py
Exercise2.pdf
exercise2.py
icdar17_labels_test.txt
icdar17_labels_train.txt
mus_run0.pkl.gz
mus_run1.pkl.gz
mus_run2.pkl.gz
mus.pkl.gz
parmap.py
skeleton__.py
skeleton.py
exercise2.py
import cv2
from parmap import parmap
from sklearn.decomposition import PCA
def parseArgs(parser):
parser.add_argument('--labels_test',
PROBLEMS DEBUG CONSOLE TERMINAL OUTPUT
TERMINAL
--in_train icdar17-historicalwi-training-color \
--labels_train icdar17_labels_train.txt \
--in_test icdar17-historicalwi-training-color \
--labels_test icdar17_labels_test.txt \
--powernorm \
--gmp \
--gamma 1.0
Command 'python' not found, did you mean:
command 'python3' from deb python3
command 'python' from deb python-is-python3
maryam@maryam:~/Downloads/Projekt Cv/Exercise 02$ python3 exercise2.py --in_train icdar17-historicalwi-training-color --labels_train icdar17_labels_train.txt --in_test icdar17-historicalwi-training-color --labels_test icdar17_labels_test.txt --powernorm --gmp --gamma 1.0
#train: 1182
> compute VLAD for test
#test: 3600
> evaluate for VLAD
Top-1 accuracy: 0.8008333333333333 - mAP: 0.5882365519967903
> compute VLAD for train (for E-SVM)
> esvm computation start
100%| | 3600/3600 [17:48<00:00, 3.37it/s]
> evaluate esvm encodings
Top-1 accuracy: 0.8008333333333333 - mAP: 0.588236551958331
maryam@maryam:~/Downloads/Projekt Cv/Exercise 02$
Ln 21, Col 23 Spaces: 4 UTF-8 CRLF {} Python 3.10.12 Quokka Prettier


Activities
Visual Studio Code
Aug 12 3:40 PM
66 %
File Edit Selection View Go Run
Exercise 02
EXPLORER
EXERCISE 02
__pycache__
icdar17-historicalwi-traini...
ScriptNet-HistoricalWI-20...
enc_test_gmp1.0.pkl.gz
enc_train_gmp1.0.pkl.gz
exercise.pdf
exercise2 (1).py
Exercise2.pdf
exercise2.py
icdar17_labels_test.txt
icdar17_labels_train.txt
mus_run0.pkl.gz
mus_run1.pkl.gz
mus_run2.pkl.gz
mus.pkl.gz
parmap.py
skeleton__.py
skeleton.py
exercise2.py
import os
import shlex
import argparse
from tqdm import tqdm
PROBLEMS DEBUG CONSOLE TERMINAL OUTPUT
TERMINAL
[WARN:0@7311.975] global loadsave.cpp:241 findDecoder imread_('./icdar17-historicalwi-training-color/icdar2017-training-color/995-IMG_MAX_116305.jpg'): can't open/read file: check file path/integrity
WARNING: cannot read image: ./icdar17-historicalwi-training-color/icdar2017-training-color/995-IMG_MAX_116305.jpg
[WARN:0@7311.976] global loadsave.cpp:241 findDecoder imread_('./icdar17-historicalwi-training-color/icdar2017-training-color/995-IMG_MAX_116306.jpg'): can't open/read file: check file path/integrity
WARNING: cannot read image: ./icdar17-historicalwi-training-color/icdar2017-training-color/995-IMG_MAX_116306.jpg
100%| | 1182/1182 [00:00<00:00, 11290.34it/s]
> multi-VLAD: computing encodings for TEST
> compute VLAD for run 1/3 (K=32)
100%| | 3600/3600 [1:21:16<00:00, 1.35s/it]
> compute VLAD for run 2/3 (K=32)
100%| | 3600/3600 [1:19:43<00:00, 1.33s/it]
> compute VLAD for run 3/3 (K=32)
100%| | 3600/3600 [1:19:03<00:00, 1.32s/it]
> PCA fit on TRAIN multi-VLAD (with whitening)
/home/maryam/.local/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:648: RuntimeWarning: invalid value encountered in divide
explained_variance_ratio_ = explained_variance_ / total_var
> evaluate multi-VLAD + PCA
Top-1 accuracy: 0.7702777777777777 - mAP: 0.5595893847937761
> E-SVM on multi-VLAD (PCA) encodings
100%| | 3600/3600 [00:53<00:00, 66.74it/s]
> evaluate E-SVM on multi-VLAD (PCA)
Top-1 accuracy: 0.765 - mAP: 0.5535101943581384
maryam@maryam:~/Downloads/Projekt Cv/Exercise 02$
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.10.12 Quokka Prettier